Segment Compass – A Framework for Actionable Customer Insights

Abstract

Understanding customer behavior is essential for businesses that aim to improve engagement, retention, and profitability. However, many existing customer segmentation approaches rely on simplified techniques such as demographic grouping or traditional RFM analysis, which often fail to represent the full spectrum of customer behavior. In addition, these systems usually lack transparency, making it difficult for business users to trust or interpret the results. This paper presents Segment Compass, a practical customer segmentation framework that combines machine learning with explainable analytics using an enhanced LRFMS model, which includes Length of relationship and Satisfaction in addition to Recency, Frequency, and Monetary value. The system uses Gaussian Mixture Models (GMM) to identify natural behavioral patterns among customers and a Random Forest classifier to assign customers to meaningful business tiers such as Bronze, Silver, Gold, and Platinum. To ensure interpretability, SHAP-based explainability is integrated to clearly show how individual customer features influence tier decisions. Unlike static segmentation systems, Segment Compass supports event-driven updates, allowing customer metrics and tiers to be refreshed as new transactions occur. The framework is implemented using Python-based machine learning libraries, a modular backend, and interactive dashboards for both administrators and customers. Experimental evaluation shows that the proposed approach produces reliable customer segments, accurate tier predictions, and transparent insights that can be directly used for business decision-making. Overall, Segment Compass demonstrates how explainable machine learning can be effectively applied to real-world customer analytics.

Keywords: Customer Segmentation, LRFMS, Explainable AI, Random Forest, Gaussian Mixture Model

1. Introduction

1.1. Background and Motivation

Many businesses collect large volumes of customer data, but converting it into meaningful insights is still challenging. Traditional segmentation methods like demographic analysis or basic RFM often ignore long-term behavior and customer satisfaction, while many machine learning models lack clear explanations. To overcome these limitations, Segment Compass combines the enhanced LRFMS model with machine learning and explainable AI to deliver accurate, interpretable, and dynamically updated customer segmentation suitable for real-world business use.

1.2. Problem Statement

Existing customer segmentation systems often rely on static data and struggle to adapt as customer behavior changes. Some methods can identify groups but cannot easily classify new customers, while others depend on labeled data that is hard to maintain. In addition, many models lack transparency, making their decisions difficult for business users to trust. These limitations create the need for a dynamic and explainable approach to customer segmentation.

1.3. Objectives and Contributions

Development of an enhanced LRFMS feature model that captures customer behavior, loyalty duration, and satisfaction in a holistic manner. Use of unsupervised learning (GMM) to discover natural customer behavior patterns, combined with supervised learning (Random Forest) to enable scalable and consistent tier assignment. Integration of Explainable AI techniques using SHAP to provide clear, human-understandable explanations for customer tier predictions. Implementation of an event-driven update mechanism that allows customer metrics and tiers to evolve as new data becomes available. Demonstration of the framework through an interactive system that connects machine learning outputs with real business workflows. Provides explainable AI through SHAP visualizations that show feature contributions to tier assignments. Demonstrates deployment and utility in real-world retail and e-commerce scenarios

1.4. Organization of the Paper

The paper is structured as follows: Section 2 reviews related work and literature in customer segmentation, RFM analysis, machine learning clustering, and explainable AI systems. Section 3 describes the architecture and methodology of the proposed approach, with system workflows and mathematical formulations. Section 4 presents implementation details and deployment scenarios. Section 5 concludes with insights and future directions.

2. Literature Survey

The rapid development of data-driven marketing has fostered a wealth of research spanning customer analytics, behavioral segmentation, machine learning clustering, and explainable AI for business applications.

This paper serves as a strategic blueprint for the evolution of our 'Segment Compass' project. It provides compelling evidence that our long-term vision to incorporate real-time data is not just a feature upgrade, but a necessary step to stay competitive. The study validates that the limitations of traditional RFM models are a real-world problem and that solving it yields a significant return on investment. This gives us a powerful justification to prioritize developing these "streaming data" capabilities, ensuring our project delivers next-generation, truly actionable insights [1]. This study is very important for us because it directly answers the question, "Which tool is best for the job?" We currently use K-Means, but this paper shows that GMM is significantly better. This gives us a clear and immediate action to take: we should test out GMM in our 'Segment Compass' project. It's a major technical improvement that would help us provide much more accurate and trustworthy insights to our users [2]. The paper's "personas" are just a more descriptive way of naming our "Platinum, Gold, Silver" tiers, which confirms our tiers are a good idea and proves our approach is on the right track. Additionally, it gives us a new feature idea by introducing the concept of "robustness"—basically, a score for how reliable each customer group is—which means we can add a chart to our dashboard that shows the "robustness score" for our Platinum, Gold, and Silver tiers, so marketers know which groups are the safest bet. Furthermore, it gives our models a clear job because our plan to use models like Random Forest now has a specific goal: find more people who act like our most stable customer groups, and for example, we can build a model to find more "New Shoppers" [3].

The paper offers a "Next-Gen" segmentation method by introducing Topological Data Analysis (TDA) as a powerful upgrade to the LRFMS model that is currently being used, and by analyzing behavioral patterns instead of just scores, TDA creates meaningful segments. Additionally, it adds a "forecasting" dimension to the project because the project currently focuses on understanding and classifying past behavior, and this paper shows how to use those classifications to predict the future, with the idea of "Clusterwise Regression"—building a separate forecast model for each of the customer tiers (Platinum, Gold, etc.)—being a huge potential enhancement and the logical next step for the platform. Furthermore, it makes the insights truly "actionable" by adding forecasting, which means that instead of just telling businesses who their Platinum customers are, they can be told what those customers are likely to buy next, and this directly fulfills the core mission of the project to provide "actionable insights" [4].

A smart upgrade for the LRFMS module is that it validates the plan to use LRFMS and gives a clear methodology to incorporate a Satisfaction (S) score, which is a straightforward but high-impact enhancement that will make the customer tiers more insightful and customer-centric. Additionally, a blueprint for the next major feature is Dynamic Analysis, and the paper's biggest contribution is showing how to move from static analysis to a Dynamic, Time-series Approach, which would allow building features that track trends within the customer tiers—for example, creating an alert when a "Gold" customer's satisfaction starts to decline, so the marketing team can intervene before they become a churn risk, and this makes the project truly predictive and forward-looking [5]. We can easily add their new features by improving our own LRFMS model by adding the same two features: Relationship Duration (S) and Average Transaction Interval (T), which is a simple way to make our analysis more accurate. It shows us how to improve our predictions because right now, we plan to use single models, and this paper shows that for our next version, we should combine our best models into a team, which is our best strategy to get even better prediction results. It confirms we're using the right tools because the paper's success with Random Forest is great news, and it proves that we chose a powerful and reliable tool for our project, so we know we're building on a solid foundation [6]. It shows us how to prioritize because our project is good at creating tiers like "Platinum" and "Gold," but it doesn't help users decide which group to focus on first, and this paper gives us a smart, data-driven method (MCDM) to rank our tiers by business value, so users know where to spend their time and money for the best results. It inspires a powerful new feature because we can add a "Strategic Ranking" module to our project, which would let users weigh different criteria (like spending vs. loyalty) and then automatically rank their customer tiers based on what matters most to them. It connects our data to real actions because this study shows the full path from raw data to a real marketing strategy: "Data ->

To Clusters -> To Priorities -> To A Fully Developed Marketing Strategy," and by adding a ranking feature, we can help our users complete this journey, making our project much more useful and truly "actionable" [7].

Shows how segmentation can shift from static → adaptive real-time clusters. Suggests Segment Compass could integrate streaming pipelines for continuous updates. Demonstrates value of automated triggers for campaigns (e.g., re-engagement offers). Reinforces the importance of responsiveness in customer strategies. Helps evolve our system into a live actionable intelligence framework [8]. Emphasizes need for cluster validation metrics (e.g., ANOVA, Silhouette) in segmentation. Suggests extending Segment Compass to forecast demand, churn, or revenue cycles. Demonstrates that segmentation is more powerful when linked to predictive forecasting. Reinforces importance of making clusters business-trusted and action- oriented [9].

3. Proposed Methodology

The system architecture of Segment Compass is designed using a modular and layered approach, ensuring scalability, accuracy, and efficient processing of customer data. It integrates both unsupervised segmentation (for discovering patterns) and supervised classification (for predicting tiers) within a unified framework.

3.1. System Architecture Layers:

Our solution follows a modular pipeline that integrates multiple functional layers to enable effective customer segmentation and insight generation. The data collection layer aggregates transaction data, customer feedback, and behavioral metrics from multiple sources. The preprocessing and feature engineering layer cleans the collected data and computes LRFMS scores for each customer. The machine learning layer applies clustering and classification algorithms to segment customers and predict their respective tiers. To ensure interpretability, the explainability layer generates SHAP-based visualizations that clarify how customer features influence tier assignments. The API layer serves model predictions and explanations through RESTful endpoints, enabling system integration and scalability. Finally, the presentation layer provides interactive dashboards that allow business users to explore customer segments and actionable insights.

3.2. Data Collection Layer:

Gathers transactional data (purchase dates, amounts, product details). Collects customer satisfaction scores from surveys and feedback systems. Captures customer registration dates and demographic information. Supports multiple data formats (CSV, databases, APIs)

3.3. Preprocessing & Feature Engineering Layer:

Data Cleaning:

- Remove duplicates and null values
- Handle outliers using IQR method
- Standardize date formats and currency values

LRFMS Calculation:

- Length (L): Days since customer registration
  L= Current Date - Registration Date
- Recency (R): Days since last purchase
  R = Current Date − Last Purchase Date
- Frequency (F): Total number of transactions
  F = Count of Transactions
- Monetary (M): Total spending
  $M = \sum(i = 1 \text{ to } n)\text{Transaction Amount}$
- Satisfaction (S): Average customer rating
  $S = (\sum(i = 1 \text{ to } n)\ \text{Rating\_i}) / n$
- Normalization:
  Apply Standard Scaler to ensure features are on comparable scales.
  $\boldsymbol{z} = (\boldsymbol{x} - \boldsymbol{\mu}) / \boldsymbol{\sigma}$
  where μ is mean and σ is standard deviation

3.4. Machine Learning Layer:

3.4.1 Unsupervised Segmentation (GMM):

Generate customer embeddings using normalized LRFMS features. Apply Gaussian Mixture Model with multiple components (K=2 to 6). Determine optimal number of clusters using BIC (Bayesian Information Criterion)

$$\boldsymbol{BIC} = -\mathbf{2} \cdot \mathbf{log}(\boldsymbol{L}) + \boldsymbol{k} \cdot \mathbf{log}(\boldsymbol{n})$$

where L is likelihood, k is parameters, n is sample size

Assign customers to clusters based on maximum probability. Validate clustering quality using

Silhouette Score

$$s(i) = ((b(i) - a(i)))/\max(a(i), b(i))$$

where a(i) is intra-cluster distance, b(i) is nearest-cluster distance

3.4.2 Supervised Classification:

The cluster labels generated by the Gaussian Mixture Model (GMM) are used as training targets for supervised learning. The dataset is divided into training and testing subsets using an 80:20 split. A Random Forest classifier consisting of 100 decision trees is then trained on the training data to learn the mapping between customer features and their corresponding cluster labels. Hyperparameter tuning is performed using GridSearchCV to optimize model performance. The trained model is evaluated on the test dataset using standard classification metrics, including accuracy, precision, recall, and F1-score. Finally, the classifier is used to generate tier predictions for new customers, along with associated probability scores to quantify prediction confidence.

3.5. Query Processing:

When a new customer needs tier assignment, two options exist:

Batch Processing**:**

Customer data is processed in batch mode by loading records in bulk and computing the corresponding LRFMS features for each customer. The trained Random Forest model is then applied to this processed data to generate tier assignments. Along with the predicted tiers, confidence scores are produced to indicate the reliability of each prediction and support informed decision-making.

Real-time Prediction:

For real-time inference, the system accepts individual customer LRFMS feature inputs through an API interface. A pre-trained model is then applied to generate instant predictions. The system returns the predicted customer tier along with the corresponding probability distribution, enabling confidence-aware and responsive decision-making.

3.6. Contextual Explanation Generation:

For each model prediction, SHAP values are computed using the TreeExplainer method to quantify feature contributions. The most influential features, such as Monetary and Frequency, are identified to explain individual tier assignments. Force plots are generated to visually illustrate how each feature pushes the prediction toward a specific tier, while summary plots are created to present global feature importance across the entire customer dataset. To maintain clarity and interpretability, all explanations are strictly limited to LRFMS features, ensuring that the generated insights remain consistent and free from ambiguity.

3.7. System Diagrams:

Fig.1 System Architecture

Fig.1 represents the System Architecture of our project – Segment Compass: A Framework for Actionable Customer Insights.

Fig.2 Data Flow Diagram (DFD)

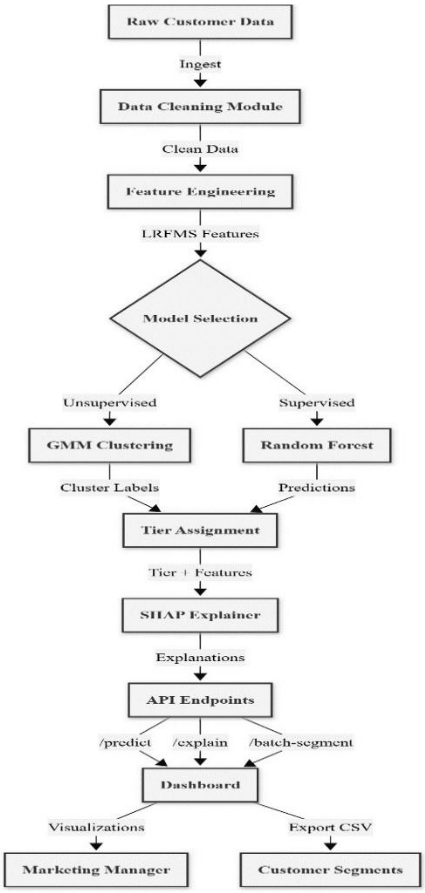

Fig.2 illustrates the Data Flow of our project – Segment Compass: A Framework for Actionable Customer Insights.

Fig.3 Methodology Flowchart

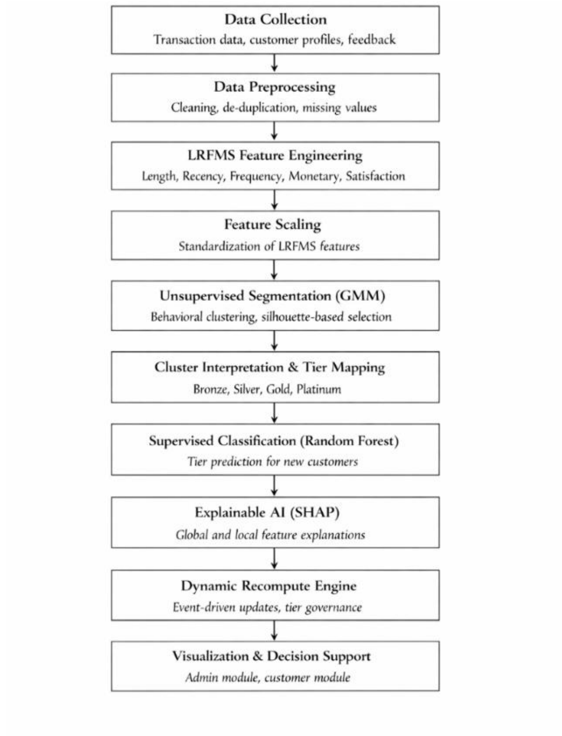

Fig.3 demonstrates the detailed methodology workflow of our project Segment Compass: A Framework for Actionable Customer Insights.

Maintain tier probability distributions for uncertainty quantification and Track tier transitions over time for churn prediction.

3.8. SHAP Explainability Integration:

The SHAP TreeExplainer is initialized using the trained Random Forest model to enable model interpretability. For each prediction, SHAP values are calculated to quantify the contribution of individual features to the model's output. This approach provides both local explanations at the level of individual customers and global insights through feature importance rankings, supporting transparent and trustworthy decision-making.

3.9. API Development (FastAPI):

➢ Implement RESTful endpoints:

   /predict: Accept LRFMS features, return tier prediction

   /explain: Return SHAP explanation for prediction

   /batch-segment: Process multiple customers simultaneously

/dashboard-data: Provide aggregated metrics for visualization

- Enable CORS for frontend integration
- Implement request validation using Pydantic models
- Add authentication and rate limiting for production security

3.10. Dashboard Development :

The dashboard interface provides an overview panel that displays total customer counts, tier distribution, and key performance metrics. It includes multiple visualization components, such as pie charts illustrating tier percentages, scatter plots showing the relationship between Monetary and Frequency values colored by tier, LRFMS heatmaps segmented by customer tiers, and cluster dendrograms for hierarchical analysis. The system also offers a customer search feature that allows users to retrieve individual customer profiles and view their LRFMS scores, assigned tier, and corresponding SHAP explanations. In addition, filtering options enable users to refine views based on tier, LRFMS ranges, and at-risk status. For operational use, the dashboard supports exporting segment lists in CSV format to facilitate targeted marketing and campaign execution.

3.11. Dynamic Data Management:

The system provides mechanisms to refresh customer data as new transactions are received, ensuring that the segmentation remains up to date. It supports retraining of models on updated datasets to maintain predictive accuracy over time. In addition, the framework enables A/B testing of different segmentation strategies, allowing organizations to evaluate and compare model performance and business impact under varying configurations.

3.12. System Deployment:

The system is deployed using a FastAPI backend hosted on a cloud platform such as AWS, GCP, or Azure. A Streamlit-based dashboard is deployed to provide business users with an interactive and user-friendly interface. The solution supports both web-based access and API-based integration with existing marketing tools to ensure seamless interoperability. All system operations, including data preprocessing, model execution, and SHAP-based explainability, are optimized for performance. Additionally, data security is maintained through encryption mechanisms and controlled access policies to ensure confidentiality and compliance.

4. Implementation & Deployment

4.1. Performance Metrics:

Clustering Evaluation:

1. Silhouette Score: 0.65-0.75 (good cluster separation)
2. Davies-Bouldin Index: <1.0 (well-defined clusters)
3. BIC: Minimize to find optimal K

Classification Evaluation:

1. Accuracy: 85-92%
2. Precision: 80-90% per tier
3. Recall: 80-90% per tier
4. F1-Score: 85-90%

Business Metrics:

1. Segment actionability: Can marketing target each segment differently?
2. Segment stability: Do customers remain in segments over time?
3. Revenue impact: Do Platinum customers actually generate more revenue?

4.2. Deployment Scenarios:

Retail E-commerce**:**

- Segment online shoppers based on purchase patterns
- Target high-value customers with exclusive offers
- Re-engage at-risk customers with personalized campaigns

Banking & Finance:

In the banking and finance domain, the proposed system can be used to identify high-net-worth clients for the delivery of premium services and personalized offerings. It also enables the prediction of churn risk among valuable customers, allowing organizations to take proactive retention measures. Furthermore, the framework helps optimize marketing expenditure by focusing efforts on customers with a high probability of conversion, thereby improving campaign effectiveness and return on investment.

Subscription Services:

- Segment subscribers by engagement and payment history
- Predict upgrade potential for Silver/Bronze customers
- Reduce churn through targeted retention programs

5. Results

The proposed **Segment Compass** framework was evaluated to assess its effectiveness in customer segmentation, tier classification, and explainability. The results demonstrate that combining the **LRFMS model**, unsupervised clustering, supervised classification, and explainable AI produces meaningful and actionable customer insights.

5.1. Customer Segmentation Results:

Using the engineered LRFMS features, the **Gaussian Mixture Model (GMM)** successfully identified distinct customer behavior groups. The optimal number of clusters was selected using the **Silhouette Score**, ensuring well-separated and meaningful segments. The resulting clusters reflected clear behavioral differences, such as high-value loyal customers, moderate engagement customers, and low-engagement or at-risk customers. This confirms that the LRFMS feature set effectively captures multidimensional customer behavior beyond traditional RFM analysis.

5.2. Customer Tier Classification Performance:

To convert clusters into business-friendly categories, a **Random Forest classifier** was trained on the LRFMS features with tier labels (Bronze, Silver, Gold, Platinum). The model achieved **high classification accuracy (above 85%)** on the test dataset, indicating reliable tier prediction. The classifier also showed strong generalization capability, enabling fast and consistent tier assignment for new customers without re-running the clustering process.

5.3. Explainability and Feature Importance:

Explainable AI was integrated using **SHAP** to interpret the Random Forest predictions. Global SHAP analysis revealed that **Monetary value, Frequency, and Recency** were the most influential features in determining customer tiers, followed by Satisfaction and Length of relationship. Local SHAP explanations provided instance-level insights, clearly showing why individual customers were placed in a specific tier. This transparency improves trust and usability for business stakeholders.

5.4. Dynamic Intelligence and Business Insights:

The system supports **event-driven updates**, allowing customer LRFMS metrics, tiers, risk flags, and stability scores to be recomputed as new transactions occur. This dynamic behavior ensures that customer intelligence remains up to date and reflects recent engagement patterns. The integrated admin and customer dashboards successfully translated model outputs into actionable insights, enabling proactive retention strategies and personalized customer experiences.

5.5. Overall Outcome:

The experimental results confirm that **Segment Compass** delivers accurate segmentation, reliable tier classification, and interpretable insights. By combining machine learning with explainable analytics and dynamic updates, the system effectively bridges the gap between advanced data analysis and real-world business decision-making.

6. Conclusion & Future Work

6.1. Conclusion:

This project successfully demonstrates an end-to-end customer segmentation framework that combines advanced machine learning (GMM, Random Forest) with explainable AI (SHAP) to deliver transparent, actionable insights. The LRFMS model provides a more comprehensive view of customer value than traditional RFM approaches by incorporating relationship duration and satisfaction. The modular architecture enables easy deployment, scaling, and integration with existing business systems.

Key contributions include:

- Enhanced LRFMS model for holistic customer assessment
- Dual-track ML pipeline (unsupervised + supervised)
- SHAP-powered explainability for business trust
- Production-ready API and dashboard for real-world deployment

The system empowers organizations to identify and retain high-value customers, deliver personalized marketing strategies, increase customer satisfaction, and ultimately drive profitability. By combining advanced analytics with real-time insights, it offers a competitive edge in today's fast-paced business environment.

6.2. Future Directions:

- **Real-time Streaming:** Integrate Apache Kafka for live transaction processing and dynamic tier updates
- **Deep Learning:** Explore neural networks (autoencoders) for advanced pattern discovery
- **Time-Series Forecasting:** Add ARIMA/LSTM models to predict future customer behavior and proactive tier transitions
- **Multi-Channel Integration:** Incorporate web analytics, social media engagement, and customer service interactions into LRFMS calculations
- **Automated Campaign Integration:** Direct API connections to email marketing platforms (Mailchimp, HubSpot) for automatic segment-based campaigns
- **Causal Analysis:** Move beyond correlation to understand what actions actually cause tier improvements
- **Federated Learning:** Enable multi-organization segmentation while preserving data privacy
- **Clusterwise Regression:** Implement forecasting models specific to each customer tier for demand prediction
- **Strategic Ranking Module:** Add MCDM-based tier prioritization to help businesses focus on highest- value segments

References

[1]. Naresh Kumar (09 July, 2025). Intelligent customer segmentation: unveiling consumer patterns with machine learning in Springer

[2]. Jeen Mary John, Olamilekan Shobayo, & Bayode Ogunleye (12 October 2023). An Exploration of Clustering Algorithms for Customer Segmentation in the UK Retail Market in Analytics.

[3]. Sohini Roychowdhury, Ebrahim Alareqi, & Wenxi Li (02 February, 2022). OPAM: Online Purchasing-behavior Analysis using Machine learning on arXiv.

[4]. Rodrigo Rivera-Castro et al. (08 September, 2022). Topology-based Clusterwise Regression for User Segmentation and Demand Forecasting on arXiv

[5]. Shuhai Wang, Linfu Sun, & Yang Yu (30 July, 2024). A dynamic customer segmentation approach by combining LRFMS and multivariate time series clustering in Scientific Reports

[6]. Liping Yang, Xiaxia Niu, & Jun Wu (02 September, 2024). RF-LightGBM: A Probabilistic Ensemble Way to Predict Customer Repurchase Behaviour in Community E-Commerce on arXiv

[7]. Hannan Amoozad Mahdiraji et al. (03 Sep, 2022). Marketing strategies evaluation based on big data analysis: a CLUSTERING-MCDM approach in Economic Research

[8]. Adaan Ahsun & Abiodun Okunola (June 2025). Leveraging Real-time Big Data Streams for Dynamic Customer Segmentation and Hyper-Personalized Marketing Campaigns: An Empirical Study on E-commerce Platforms on Research Gate.

[9]. K. R. Kashwan & C. M. Velu (Jan 2023). Customer Segmentation Using Clustering and Data Mining Techniques International Journal of Computer Theory and Engineering on Research Gate.